

2024



Fireworks AI

Fireworks.ai Delivers 4x Throughput for Generative AI and Cuts Latency by up to 50% Using AWS and NVIDIA

Learn how Fireworks.ai built a cost-optimized generative AI solution using Amazon EC2 P5 Instances powered by NVIDIA H100 Tensor Core GPUs.

[Overview](#) | [Opportunity](#) | [Solution](#) | [Outcome](#) | [AWS Services Used](#)

4x higher throughput

per instance than open-source solutions

Up to 50%

cut in latency

4x reduction

in overall costs for some customers

Simplified

developer access to large language models



Hi, I can connect you with an AWS representative or answer questions you have on AWS.



Need more info? Highlight any text to get an explanation generated with AWS generative AI.



[Fireworks.ai](#) set out to build a lightning-fast, affordable, and customizable generative artificial intelligence (AI) inference solution for its customers. With billions of parameters, foundation models require powerful, often costly compute resources as they're put into production. The company's founders sought to make these foundation models widely available for developers to incorporate into their applications while keeping costs reasonable for customers, and Fireworks.ai turned to Amazon Web Services (AWS).

Fireworks.ai powers its solution using specialized instances from [Amazon Elastic Compute Cloud](#) (Amazon EC2), which provides secure and resizable compute capacity for virtually any workload. It upgraded to [Amazon EC2 P5 Instances](#) powered by NVIDIA H100 Tensor Core GPUs, which are the highest-performance GPU-based instances for deep learning and high-performance computing applications. Fireworks.ai's generative AI solution delivers four times higher throughput per instance than open-source solutions, cuts latency in half for some customers, and meets strict enterprise-level security standards.



Opportunity | Using Amazon EC2 P5 Instances to Build a Flexible, Cost-Optimized Generative AI Solution for Fireworks.ai

The CEO and cofounder of Fireworks.ai, Lin Qiao, had led the development and productionization of the open-source, popular deep learning framework PyTorch while she served as senior director of engineering at Meta. In fact, many in the Fireworks.ai founding team worked on PyTorch to empower developers in the building of fast, secure, affordable generative AI products. "Achieving optimal cost-performance for scale and productionization is a primary challenge for customers developing on PyTorch," says Dmytro Dzulgakov, Fireworks.ai cofounder and chief technology officer. "It's particularly true with generative AI products and models because of their sheer size as well as how new and fast this field is. We wanted to use AWS to help to bridge this gap."

Fireworks.ai's inference solution both hosts generative AI software-as-a-service and supports deployments in a customer's virtual private cloud. Developers can choose to use state-of-the-



language, image, and multimodal foundation models off the shelf, or they can customize and fine-tune models. For example, Fireworks.ai supports Stable Diffusion XL, a deep learning, text-to-image model; Llama 2 large language models with up to 70 billion parameters; and StarCoder, designed specifically for code-related tasks.

Solution | Helping Customers Triple Traffic per Instance as They Lower Costs 4x

From its inception, Fireworks.ai has collaborated closely with AWS teams to determine how best to optimize compute power for its highly demanding inference engine. At first, the company used [Amazon EC2 P4d Instances](#), powered by NVIDIA A100 Tensor Core GPUs, which offer high performance for machine learning training and high-performance computing applications in the cloud. In July 2023, AWS announced Amazon EC2 P5 Instances, the next generation of Amazon EC2 P4 instances and the highest-performance GPU-based instances for deep learning and high performance computing applications. “AWS has the latest and greatest hardware,” Dzhulgakov says.

Using Amazon EC2 P5 Instances, Fireworks.ai meets demanding customer requirements for performance. For example, one customer lowered the latency of its summarization model—which uses natural language processing techniques to condense written content—by 30–50 percent. “We support about three times higher traffic with a single instance,” says Dzhulgakov. “Even though the instance itself is more expensive if you calculate the cost per single request, there actually ends up being overall cost improvement.” In fact, the customer cut total costs by four times.

Another Fireworks.ai customer, Sourcegraph, uses StarCoder, which runs on Fireworks.ai, to power inference and output for the open-source edition of its AI coding assistant, Cody. After Fireworks.ai adopted Amazon EC2 P5 Instances, Cody doubled its completion acceptance rate, a measure of a model’s output efficiency. Additionally, Cody’s backend latency accelerated by more than two times. “Using AWS, Fireworks.ai helps developers integrate powerful open models into their prototype applications without breaking the bank as they experiment, explore, and play with different models,” Dzhulgakov says. “As their product grows and usage increases, the focus shifts toward speed and cost. Using Amazon EC2 P5 Instances, we provide outstanding cost per performance for our customers’ use cases.”

And because the Fireworks.ai inference solution built on AWS is HIPAA and SOC2 Type II compliant, it meets important considerations for companies with strict security requirements for their data. “Using AWS infrastructure provider does help a lot in achieving certification for our solution,” says Dzhulgakov.



Outcome | Providing Excellent Cost per Performance for Customers' Use Cases

In addition to receiving technical guidance, Fireworks.ai has collaborated with the AWS team on marketing opportunities. Fireworks.ai customers can simplify deployment and billing by purchasing Fireworks.ai's [generative AI platform-as-a-service](#) through [AWS Marketplace](#), a curated digital catalog that customers can use to find, buy, deploy, and manage third-party software, data, and services to build solutions and run their businesses.

Fireworks.ai aims to add more multimodal foundation models to its solution so that customers can implement function-calling, invoke other external tools, and interact with their applications in richer ways.

"It primarily comes down to the accessibility of the latest hardware, making sure that it works reliably and scales to capacity needs as our customers grow and as we grow," Dzhulgakov says. "Those are the major highlights of working alongside AWS."

About Fireworks.ai

Founded in 2022, Fireworks.ai provides a fast, affordable, and customizable solution for generative artificial intelligence that helps product developers run, fine-tune, and share large language models.

AWS Services Used

Amazon Elastic Compute Cloud (Amazon EC2)

Amazon Elastic Compute Cloud (Amazon EC2) offers the broadest and deepest compute platform, with over 750 instances and choice of the latest processor, storage, networking, operating system, and purchase model to help you best match the needs of your workload.

[Learn more »](#)

Amazon EC2 P5 Instances

Amazon Elastic Compute Cloud (Amazon EC2) P5 instances, powered by the latest NVIDIA H100 Tensor Core GPUs, deliver the highest performance in Amazon EC2 for deep learning (DL) and high performance computing (HPC) applications.

[Learn more »](#)



More Software & Internet Customer Stories

GERMANY



HeyJobs on AWS

HeyJobs aims to make job searching and recruitment more effective for candidates and employers.

2024

ITALY



Namirial Streamlines Contract Signing with...

Namirial enhanced its eSignAnywhere solution with generative AI using Amazon Bedrock, improving engineering productivity and the user

2024

COSTA RICA



Modernizing Applications in Minutes Using Amazon Q...

Learn how Novacomp accelerated

2024

AUSTRALIA



Dovetail Enhances Customer Insights with...

Learn how Dovetail accelerates development of new generative AI products and features using Amazon Bedrock, deploying the latest large language models while reducing security and administrative overhead.

2024



Get Started

Organizations of all sizes across all industries are transforming their businesses and delivering on their missions every day using AWS. Contact our experts and start your own AWS journey today.

Contact Sales

Learn About AWS

- What Is AWS?
- What Is Cloud Computing?
- AWS Accessibility
- AWS Inclusion, Diversity & Equity
- What Is DevOps?
- What Is a Container?
- What Is a Data Lake?
- What is Artificial Intelligence (AI)?
- What is Generative AI?
- What is Machine Learning (ML)?
- AWS Cloud Security
- What's New
- Blogs
- Press Releases

Resources for AWS

- Getting Started
- Training and Certification
- AWS Solutions Library
- Architecture Center
- Product and Technical FAQs
- Analyst Reports
- AWS Partners

Developers on AWS

- Developer Center
- SDKs & Tools
- .NET on AWS
- Python on AWS
- Java on AWS
- PHP on AWS
- JavaScript on AWS

Help

- Contact Us
- Get Expert Help
- File a Support Ticket
- AWS re:Post
- Knowledge Center
- AWS Support Overview
- Legal
- AWS Careers



Amazon is an Equal Opportunity Employer: *Minority / Women / Disability / Veteran / Gender Identity / Sexual Orientation / Age.*

Language

- عربي |
- Bahasa Indonesia |
- Deutsch |
- English |



Español |
Français |
Italiano |
Português |
Tiếng Việt |
Türkçe |
Русский |
ไทย |
日本語 |
한국어 |
中文 (简体) |
中文 (繁體)

Privacy
|
Accessibility
|
Site Terms
|
Cookie Preferences
|

© 2024, Amazon Web Services, Inc. or its affiliates. All rights reserved.

